Matthew Shiley HFCV Summer 2025 Plan & Notes

Plan

June 2025						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	**19** **Juneteenth**	20	21
22	23 Start date Make plan Inventory	24 Machine linkages	25 J gone, assemble/ do computer stuff	26 Machine linkages Work on 3D model	27 Busy	28
29	30 Work on 3D model					

July 2025						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		1 Machine linkages	2	3	**4** **Independence Day**	5
6	7 Busy	8 Flight canceled, couldn't get back	9 Linkages	10 Linkages	11 Linkages	12
13	14 Linkages (hopefully will finish lathing today)	15 Linkages (finish milling?)	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

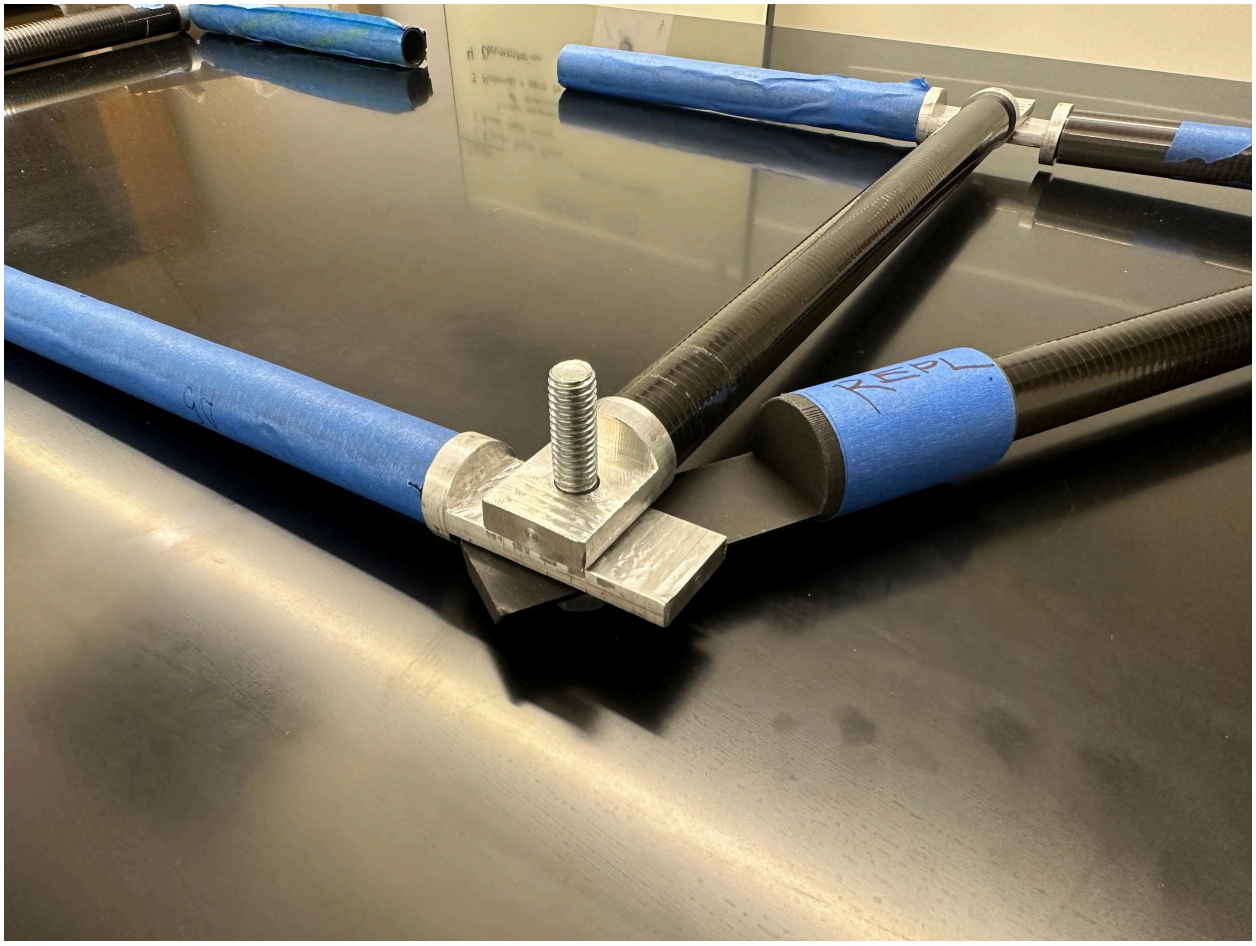
August 2025						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					1 Last day in person	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

Notes

Week 1: 6/23-6/26

Mon 6/23

- Met with Prof. Everbach to check in, layout direction for project.
- Checked in with J
    - J gone on wed 6/25
    - Need to familiarize myself with the lathe here
    - Plan on machining linkages tue 6/24 and thu 6/26
- Hope to finish linkages by next week so I can start assembly of the front frame
- Need more bolts



- Getting this three-member joint to lay flat seems like may be an issue. I should make sure everything is in line before I start making more linkages.
    - Nora informed me that washers were used before, but I think it would be good to have a design that doesn't need them.

- I decided to make a model of the front section in Fusion to help me to figure out how to alter the linkage:



- This was a good exercise as it was my first time modeling with multiple components. I will ask Nora if turning this into a full model of the car is of interest. If so, I may work on this when I don't have mechanical work to do.

- Instead of altering the 3D printed part, I filed down a side of the linkage so that the whole joint lies flat:



- Need for front:
    - 18 end members

    - 2 two-sided members (or three if the off-center hole is a problem)

- Need for back:

Tue 6/24

- Spent the day machining stopper linkages in the shop.
    - For the first couple hours, I familiarized myself with the lathe and other tools
- I managed to make 7, 9 more to go.

- I went up to SH124 to assemble as much of the chassis as I could:



Wed 6/25

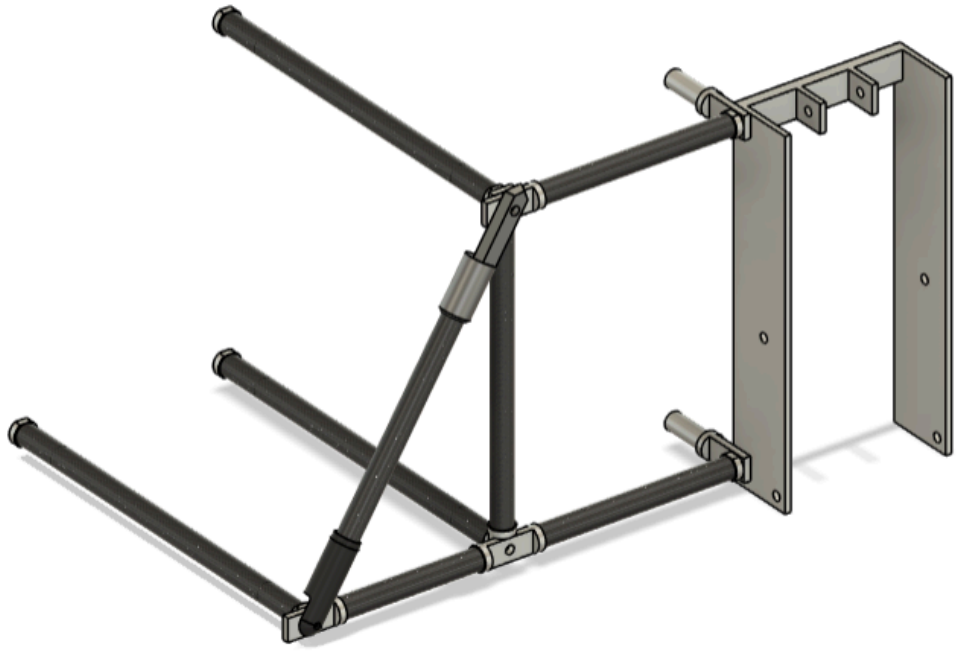
- Machined more stopper linkages in the shop. Managed to make five more.
    - Discovered that the threads needed to be tapped deeper, and the ½” relief at the top had to be bored out from ⅜” to 25/64”. This way, the tap was able to make it through.
- Attended pizza lunch & talk.

- Assembled more of the chassis with new linkages; screws still not going in far enough after tapping deeper. Not sure why.

Thu 6/26

- I decided to take a break from machining linkages because it was getting very tiresome.
- I re-assembled part of the chassis with the new linkages:

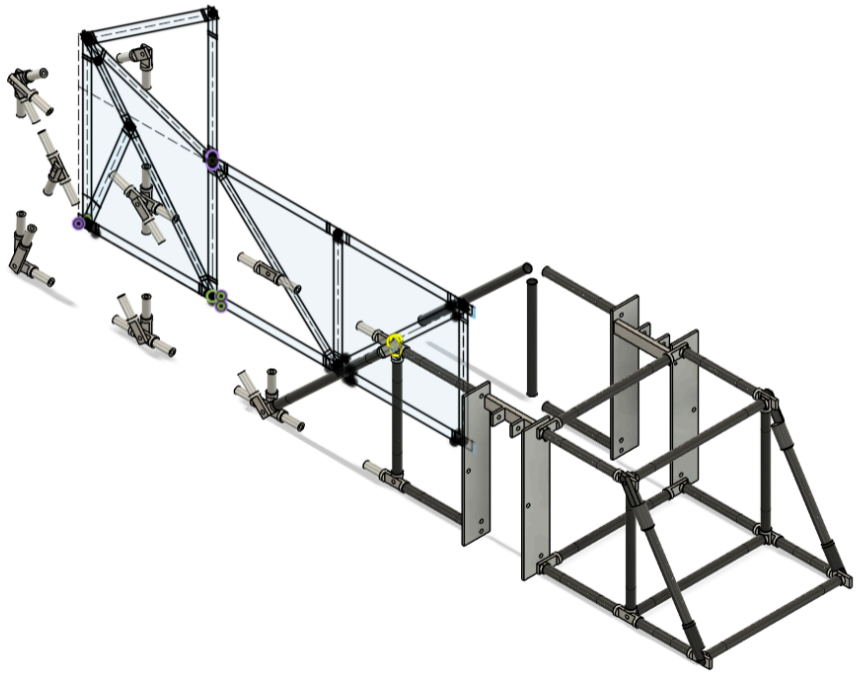
- For the rest of the day, I continued to work on the Fusion model of the car. There were several hiccups that I still haven't found solutions to.



Week 2: 6/30-7/3

Mon 6/30

Made significant progress with the 3D model of the car and verifying how many of each type of linkage I need to make.



Week 3: 7/9-7/11

Wed 7/9

Still need to make:

- 12 x Stoppers
- 3 x suspension stoppers
- 3 x 1” single
- 6 x 1.5” single
- 2 x 1.5” single offset
- 2 x 1.75” single
- 7 x 2” double
- 2 x 2.3” single
- 2 x 2.3” single offset
- 1 x 2.3” double(side)
- 4 x 2.6” single

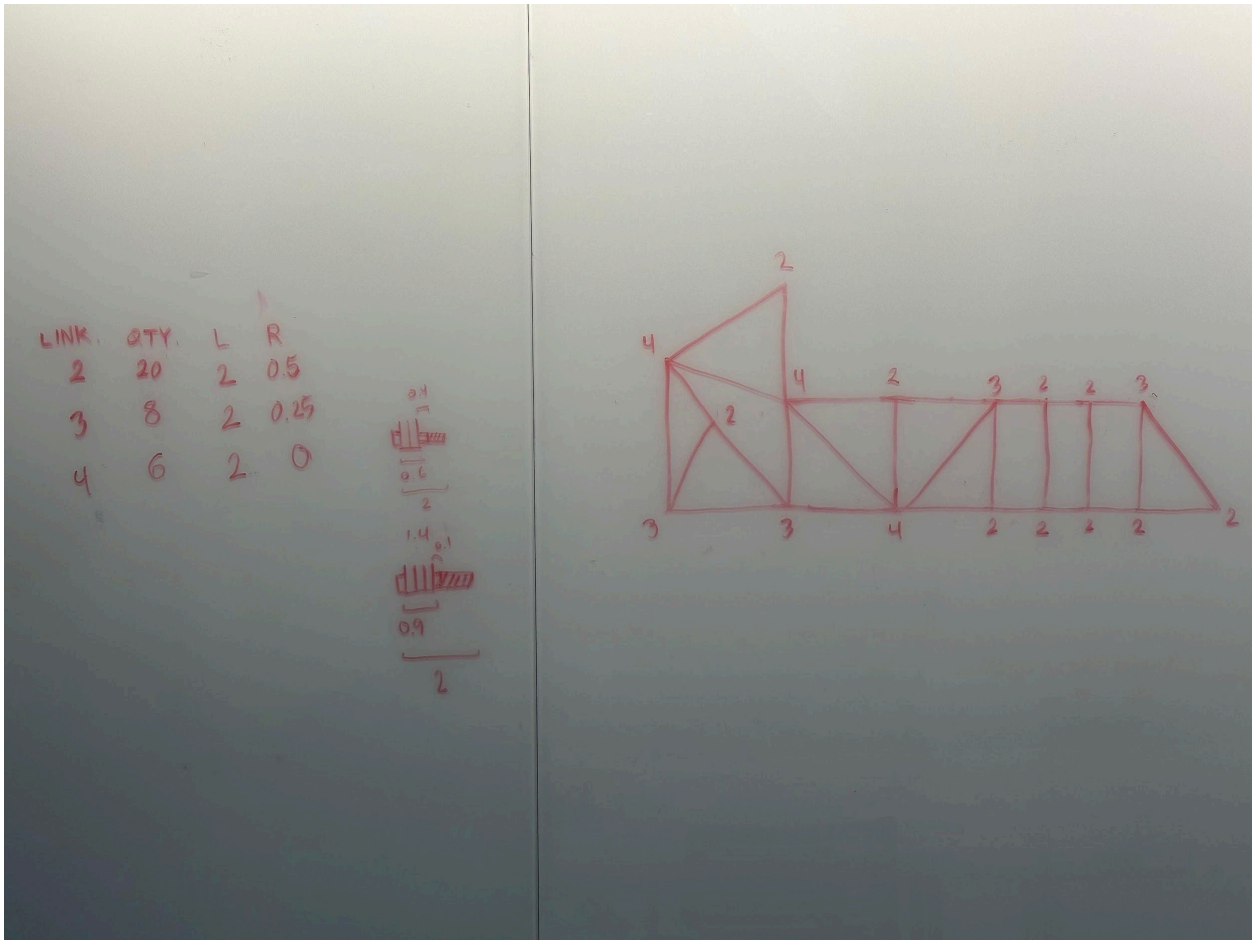
Fri 7/11

Still need:

- 2 x stoppers
- 7 x suspension stoppers
- 3 x 1” single
- 2 x 1.5” single offset
- 2 x 1.75” single
- 7 x 2” double
- 2 x 2.3” single
- 2 x 2.3” single offset
- 1 x 2.3” double(side)
- 4 x 2.6” single
- 1 x 2.8” double (center)
- 1 x 2.8” double (side)

Finished lathing almost all linkages. Still need:

- 2 x 2.3” single
- 2 x 2.3” single offset
- 1 x 2.3” double(side)
- 4 x 2.6” single
- 1 x 2.8” double (center)
- 1 x 2.8” double (side)



4 x no relief
4 x ¼” relief
2 x ½” relief

4 x 1.5” single
1 x 1.5” offset
1 x suspension stopper